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Enhanced piezoelectric properties and thermal stability in Mo/Cr co-doped $CaBi_2Nb_2O_9$ high-temperature piezoelectric ceramics

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ABSTRACT

Mo/Cr modified $CaBi_2Nb_{2-x}(Mo_{2/3}Cr_{1/3})_xO_9$ ($x = 0.03$, 0.04, 0.05, 0.06, 0.07, 0.075, and 0.1) were synthesized by the conventional solid-state reaction method. The effects of Mo/Cr substitution for Nb on structure, dielectric properties, piezoelectric properties and electrical conduction behaviours of CBN were investigated in detail. Results showed that moderate substitution of Nb with Mo/Cr optimized crystal structure and led to a reduction in oxygen vacancies, which effectively improved the electrical properties of the CBN ceramics. The $CaBi_2Nb_{1.93}(Mo_{2/3}Cr_{1/3})_{0.07}O_9$ ceramic showed optimal properties with a high d_{33} of ~15 pC/N along with a high T_C of ~939 °C, an enhanced electrical resistivity (ρ) of 3.33×10^5 Ω cm accompanied by a decreased dielectric loss of tanδ = 6.5% at 600 °C. Furthermore, the ceramic also exhibited a remarkable thermal stability performance that d_{33} remained 93% of its original value when the annealing temperature rose up to 850 °C. All these properties demonstrated the great potentials of Mo/Cr doped CBN-based ceramic for high-temperature applications.

1. Introduction

With the development of aircraft, aerospace and nuclear power industries, the demand for high-temperature (>600 °C) piezoelectric devices has dramatically increased, such as accelerometers and sensors used to detect and record dynamic mechanical parameters at harsh conditions [1–3]. Considering the usage temperature upper limit of lead zirconate titanate (PZT) is only about 320–430 °C [3], it is urgently needed to find alternatives that can be used in much higher temperature range. Bismuth-layered structure ferroelectrics (BLSFs) with formula of $(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$ have attracted great attention because of the high Curie temperature [4,5]. Among them, $CaBi_2Nb_2O_9$ is considered to be a particularly important member with remarkable Curie temperature of 940 °C, which might be used in high temperature environment beyond 600 °C [6]. However, the d_{33} value of pristine CBN ceramic is low (~5 pC/N), which greatly limits its practical application [7].

Several approaches have been developed to promote the piezoelectric properties of CBN-based ceramics [8–14]. Thereinto A-site doping is considered as a valid approach to tailor the crystalline structure of CBN, which could effectively improve the piezoelectric activity. Previously, d_{33} values of 13.3, 16, 16 pC/N have been reported in Li/Ce, Na/Ce, K/Ce co-modified CBN ceramics [8–10]. Furthermore, several studies also revealed that suitable A-site triatomic doping could effectively tailor the pseudo-morphotropic phase boundary, which was helpful to improve the piezoelectric properties of CBN-based ceramics. With A-site (X, Li, Ce) doping, CBN-based ceramics showed better d_{33} of 13.1and 15.3 pC/N (X = Nd and Bi, respectively) [11,12]. Moreover, Liu et al. recently found that A-site Li/Bi doping and $(Bi_2O_2)^{2+}$ layers Ce doping in CBN ceramic can manipulate the pseudo-morphotropic phase boundary. A remarkable piezoelectric constant of (20.2 pC/N) and a good thermal depoling performance (d_{33} remained 89.6% of its initial value after depoling at 875 °C) were achieved [13]. Unfortunately, only a few successful attempts on B-site modification of CBN ceramics were reported, which might be ascribed to the fact that most of the cations that could enter the B-site have similar sizes and may not play a major structural role in the polarisation process for BLSFs [15].

Recently, B-site co-doping strategy with mixed-valence cations was developed to enhance the piezoelectric properties of some BLSF ceramics. For instance, Li et al. developed a remarkable d_{33} value of ~38 pC/N in B-site Nb/Cu co-doped $Bi_4Ti_3O_{12}$ ceramics, which was the

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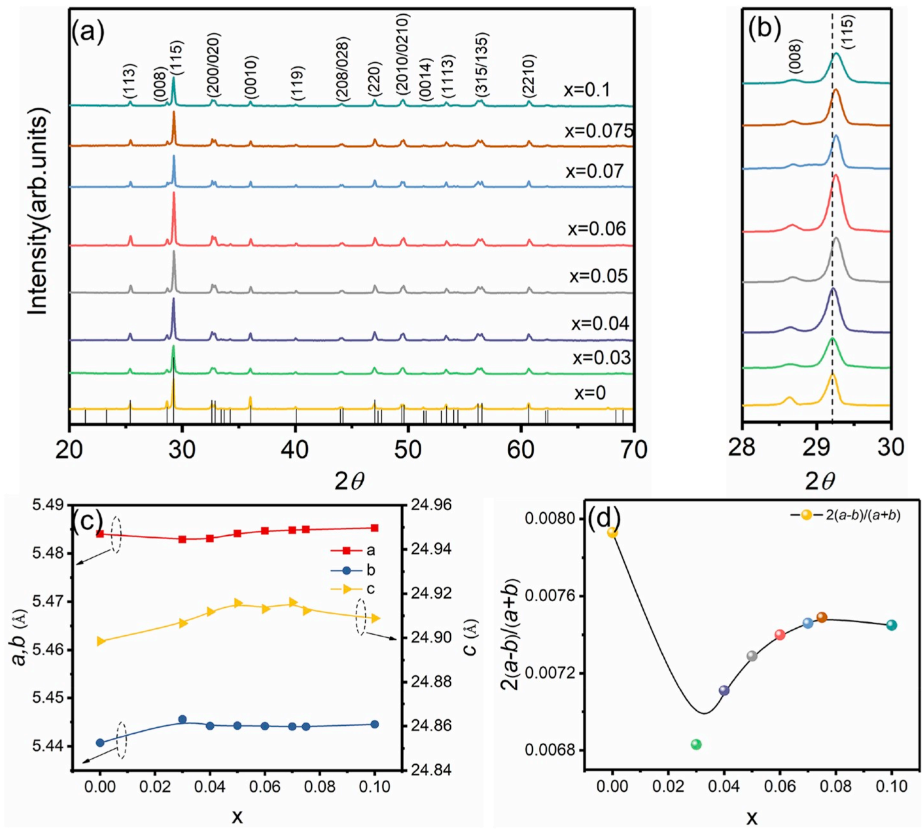

Fig. 1. (a) XRD patterns of the Mo/Cr doped CBN ceramics. (b) Magnified (115) diffraction peaks. (c) Lattice parameters *a*, *b* and *c*. (d) *2(a-b)/(a+b)* values of the Mo/Cr doped ceramics.

maximum value among the currently reported BIT ceramics [16]. Moreover, Shen et al. found a remarkable d_{33} value of 27 pC/N in Nb^{5+} and Mn^{4+} co-doped $CaBi_4Ti_4O_{15}$ ceramics obtained by a two-step sintering process [17], which is also the maximum value of CBT-based ceramics. All these studies demonstrate that the piezoelectric activity of BLSFs can be effectively enhanced by co-doping of mixed-valence cations at the B-site. Especially, in our recent work, we fabricated CBN ceramic with a high d_{33} (15 pC/N) together with a remarkable resistivity (4.9×10^5 Ω•cm at 600 °C) by introducing W/Cr at B-sties [18]. Although, the obtained d_{33} was not the highest in CBN-based ceramics, the optimized piezoelectric properties and enhanced resistivity still imply the great potentials of B-site mixed-valence cations co-doped CBN ceramics. Nevertheless, researches on CBN-based ceramics modified by B-site co-doping strategy with mixed-valence cations are still very rare until now.

Considering having the similar electronic configuration with W^{6+}, mixed-valence cations substitution for Nb^{5+} with Mo^{6+} and some low valence cations may be also valid to enhance the piezoelectric activity of CBN-based ceramics. In this work, B-site mixed-valence cations co-modified $CaBi_2Nb_{2\text{-}x}(Mo_{2/3}Cr_{1/3})_xO_9$ ceramics were synthesized by the solid-state sintering method. High d_{33} accompanied by high T_C, excellent thermal stability and high resistivity was achieved. The relationship between structure and electrical properties of Mo/Cr doped ceramics were investigated in details.

2. Experimental

$CaBi_2Nb_{2\text{-}x}(Mo_{2/3}Cr_{1/3})_xO_9$ (when $x = 0.030$, 0.040, 0.050, 0.60, 0.070, 0.075 and 0.1) ceramics were fabricated using the solid-state reaction method. As raw materials, $CaCO_3$ (99.99%), Bi_2O_3 (99.5%), Nb_2O_5 (99.5%), MoO_3 (99.95%), and Cr_2O_3 (99.95%) were weighed according to the stoichiometric compositions. All the powders were mixed in a ball mill for 12 h using ethanol as solvent, then dried and calcined at 750 °C for 2 h. The calcined powders were milled again under the same conditions. These powders were granulated by adding polyvinyl alcohol (PVA), then pressed into disks with a diameter of 12 mm and a thickness of 1 mm at a pressure of 200 MPa. After burning out PVA at 550 °C, the pellets were sintered at 1175 °C for 3 h.

X-ray diffraction (XRD) date of $CaBi_2Nb_{2\text{-}x}(Mo_{2/3}Cr_{1/3})_xO_9$ ceramic powders were collected by a Ultima IV XRD diffractometer. The microstructure characterization of the sintered disks was conducted by a JEOL JSM6460-LV Scanning Electron Microscopy. Raman scattering of the ceramics were analysed by a RENISHAW inVia Raman spectrometer. Silver paste was coated on the two surfaces as electrode layers to measure electrical properties. The relative permittivity and loss tangent of the disks were investigated at 1 MHz using a Keysight E4990A impedance analyser and the resistivity was examined from 200 °C to 800 °C using a Keithley 2410 high-resistance meter under a constant voltage of 10 V. Ferroelectric properties were ascertained by using an aixACCT TF200 ferroelectric analyzer. The specimens were performed under a DC electric field of 14–16 kV/mm at 160 °C in silicone oil for 30 min and the piezoelectric coefficients (d_{33}) were examined using a piezo-d_{33} m (ZJ-3AN) at room temperature. In order to test the thermal stability of the samples, the poled disks were annealed for 2 h from room temperature to 1000 °C and the d_{33} values were measured at room temperature.

3. Results and discussion

XRD patterns of $CaBi_2Nb_{2\text{-}x}(Mo_{1/3}Cr_{2/3})_xO_9$ ceramic powders are exhibited in Fig. 1(a). Every distinct diffraction peak matched well with the standard PDF Card (JCPDS 49-0608) of CBN ceramic. The strongest reflection plane of all samples could be assigned to the (115) plane, which was in accordance with the fact that ($112m + 1$) is the most intense reflection of BLSFs [19,20]. No trace of any impurity could be

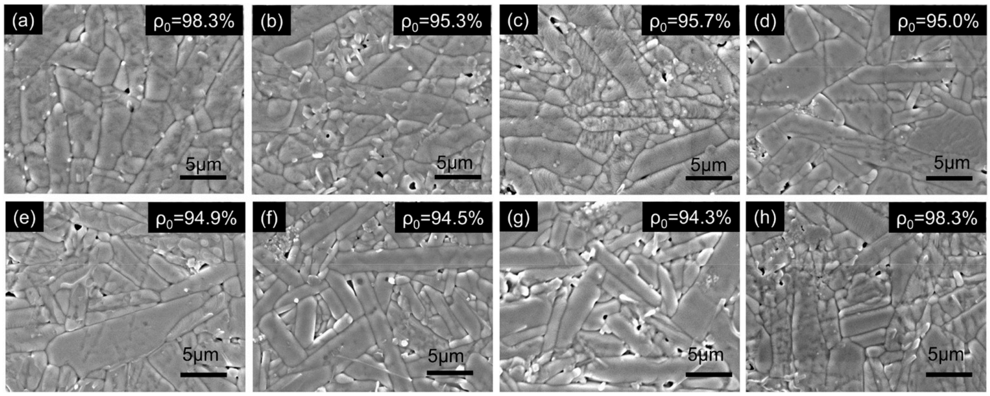

Fig. 2. SEM image of the polished thermally-etched surfaces for Mo/Cr doped ceramics: (a) $x = 0$, (b) $x = 0.03$, (c) $x = 0.04$, (d) $x = 0.05$, (e) $x = 0.06$, (f) $x = 0.07$, (g) $x = 0.075$ and (h) $x = 0.1$.

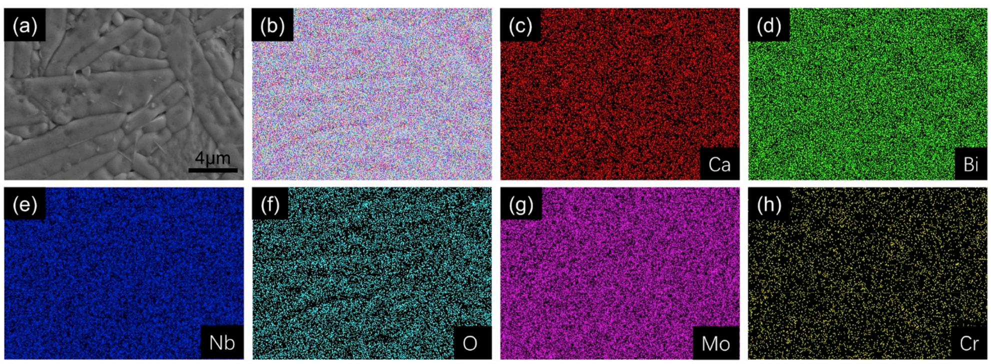

Fig. 3. Elemental mapping of the polished thermally-etched surfaces for Mo/Cr doped ceramic with $x = 0.07$.

found after the Mo/Cr doping, implying that the doped elements have totally incorporated into the crystal lattices of CBN. According to Fig. 1(b), with an increase in x, the (115) peak shifted to higher angles, implying a shrinkage of the crystal structure. The smaller average ionic radii of Mo/Cr ions (~0.57 Å) than Nb ions (~0.64 Å) may be the cause of this evolution [21–23]. Although subtle structural variation could be observed, no phase boundary was detected within the doping range.

To further elucidate the variation of crystal structure with different Mo/Cr content, lattice parameters a, b, c are illustrated in Fig. 1(c). As shown, the lattice constant a decreased while b slightly increased with an increase in x from 0 to 0.03. When $x > 0.03$, a slightly increased together with b decreased with the increasing x. The variation of $2(a\text{-}b)/(a+b)$ value which can be used to evaluate the tetragonal distortion degree is shown in Fig. 1(d). One can see that the value of $2(a\text{-}b)/(a+b)$ decreased firstly and then increased with the increasing doping level. All the doped ceramics exhibited smaller $2(a\text{-}b)/(a+b)$ value than that of undoped one, suggesting that the substitution of Mo/Cr for Nb might cause a relaxation of the tetragonal distortion [24].

The polished thermally-etched surface microstructures for Mo/Cr doped ceramics are shown in Fig. 2. All the samples exhibited plate-like grains with random orientations because of the higher grain growth rate along the direction perpendicular to the c-axis of pseudo-perovskites. No obvious change can be observed in the grain morphology, suggesting that the Mo/Cr doping slightly affected the grain growth of the CBN ceramic. Still some pores can be observed in the Mo/Cr co-doped ceramics, all the samples exhibited a relative density higher than 94%. Elemental mappings of the polished thermally-etched surface of $x = 0.07$ Mo/Cr doped ceramic are given in Fig. 3. As shown, all the elements were evenly distributed in the ceramics, demonstrating the diffusion of Mo/Cr ions into the lattice.

Fig. 4 shows the temperature dependence of dielectric constant and loss tangent of Mo/Cr doped CBN ceramics tested at 1 MHz. One can see that all the ceramics exhibited obvious dielectric peak near 930 °C, corresponding to the phase transition from orthorhombic structured

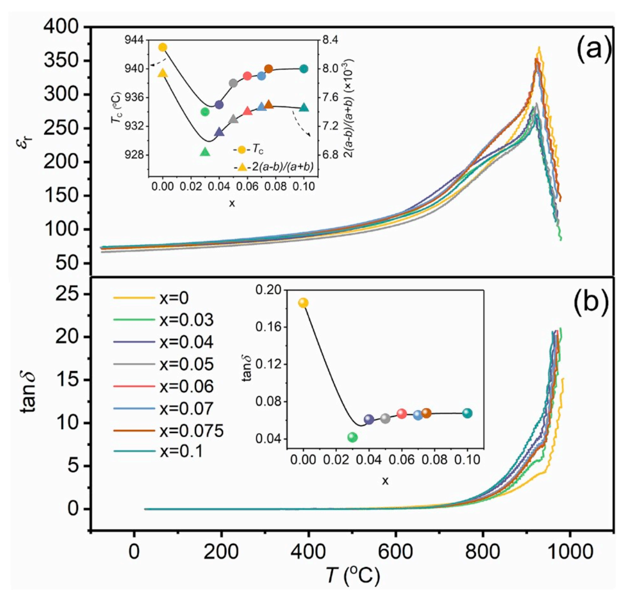

Fig. 4. Temperature dependence of the dielectric permittivity ε_r and dielectric loss $\tan\delta$ of the Mo/Cr doped CBN ceramics at 1 MHz. (a) Dielectric permittivity, with an *inset* showing the change trends in T_C and structural orthorhombicity $2(a\text{-}b)/(a+b)$ of the samples with an increase in the x value. (b) Dielectric loss, with an *inset* of $\tan\delta$ of the Mo/Cr doped CBN ceramics at 600 °C.

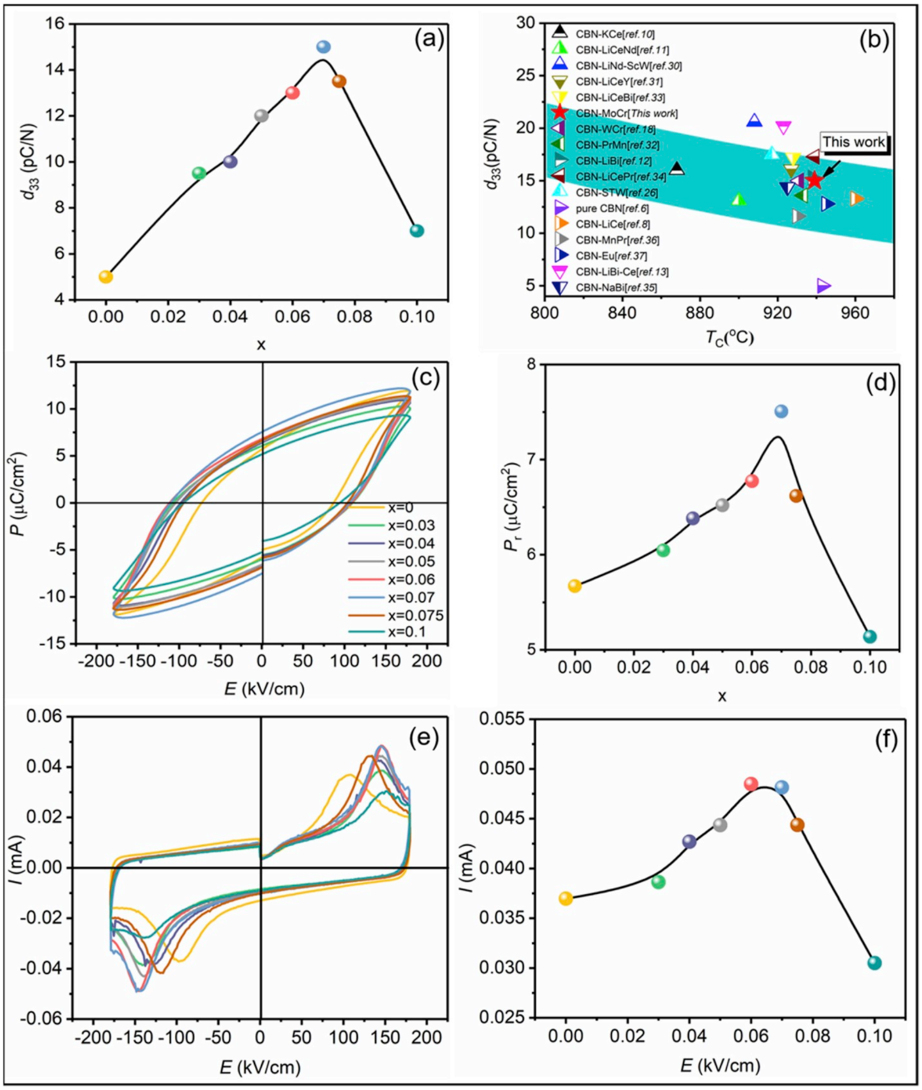

Fig. 5. (a) d_{33} values of the Mo/Cr doped ceramics as a function of x. (b) Correlation between the T_C and d_{33} of previously reported CBN ceramics. (c) *P–E* loops of the Mo/Cr doped ceramics. (d) P_r values of the Mo/Cr doped ceramics as a function of x. (e) *I-E* loops of the Mo/Cr doped ceramics. (f) *I* value of the Mo/Cr doped ceramics as a function of x.

ferroelectric phase to tetragonal structured paraelectric phase [12]. Moreover, a dielectric anomalous weak peak can be observed at temperature around 700 °C, which might be attributed to the defect related dielectric response. Similar dielectric anomaly has been observed in some other reported BLSFs ceramics [25,26]. The variation trend of T_C first declined and after that went up, as exhibited in the *inset* of Fig. 4(a). Generally, the variation of the T_C was considered to be closely related to the variation of tolerance factor (t). It has been revealed in earlier studies that smaller t value always corresponds to larger T_C in BLSFs [27]. The formula of tolerance factor (t) was given as followed:

$$t = \frac{(r_A + r_O)}{\sqrt{2}(r_B + r_O)} \tag{1}$$

where, r_A, r_B, and r_O corresponds to the ionic radii of the A-site cations, B-site cations, and oxygen ion, respectively [28,29]. Considering the smaller average ionic radii of Mo/Cr ions (~0.57 Å) compared to that of Nb ions (~0.64 Å), co-doping of Mo/Cr would lead to a continuous increase in t value and thus give a gradually decreased T_C with the increasing doping level, which was clearly not in accordance with the experimental results and suggested that the variation of tolerance factor cannot fully reflect the evolution of the lattice structure. Alternatively, from Fig. 4(a) we can see that the variation of T_C was similar to the variation tendency of $2(a\text{-}b)/(a+b)$, indicating the Curie temperature variation may be ascribed to the change of tetragonal distortion. Similar result has also be observed in previous report [13]. Besides, the loss tangent of all doped CBN ceramics (about 6%) were only one third of that of pristine one (18%) at 600 °C, which may be caused by a decrease in oxygen vacancy concentration after Mo/Cr doping. Detailed mechanism will be discussed later. The undeteriorated T_C along with low tanδ demonstrated that Mo/Cr modified CBN ceramics could be a potential candidate in high temperature application.

Fig. 5(a) depicts the d_{33} of the Mo/Cr doped ceramics. As shown, the d_{33} value increased firstly and then decreased as the x increased. All doped samples exhibited larger d_{33} value than the pristine one, indicating that the piezoelectricity of CBN ceramics could be effectively improved by introducing Mo/Cr substitution for Nb. The ceramic with $x = 0.07$ exhibited the maximum d_{33} value of 15 pC/N. A summary of literature data relating to the correlation of T_C and d_{33} of CBN-based ceramics is shown in Fig. 5(b) [8,10–13,18,26,30–37]. It can be found that the d_{33} of the Mo/Cr doped CBN ceramic was comparable to most of the A-site doped ones, suggesting that strong piezoelectric activity can be obtained by B-site mixed-valence cations doping. The improvement in piezoelectricity is generally related to the variation in ferroelectricity.

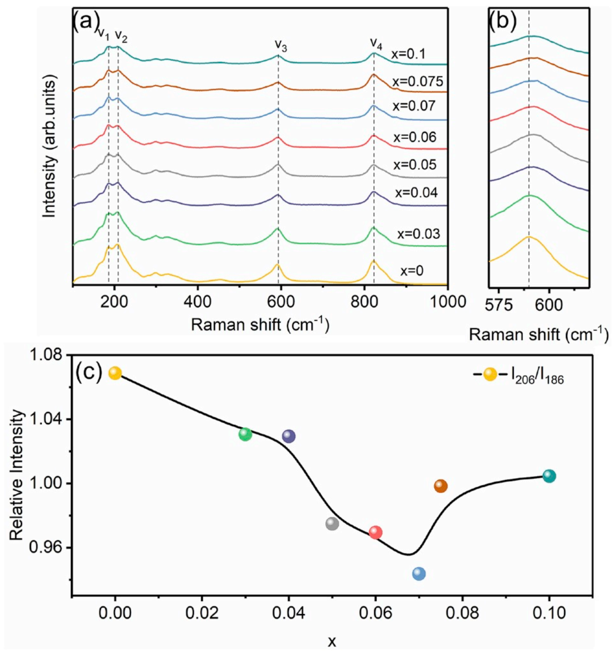

Fig. 6. (a) Raman spectra of the Mo/Cr doped ceramics at room temperature. (b) Selected regions of the peak at about 590 cm^{-1}. (c) The normalized intensity of 206 cm^{-1} peak with respect to the 186 cm^{-1} peak.

Fig. 5(c) illustrates the *P-E* loops of Mo/Cr doped ceramics tested at 1 Hz and 100 °C. It could be observed that all the Mo/Cr modified CBN ceramics displayed typical ferroelectric *P-E* loops, while the undoped one exhibited a relatively flat *P-E* loop. The P_r values *vs* x are displayed in Fig. 5(d). The P_r increased firstly and then decreased when x increased, peaked at $x = 0.07$ (7.5 μC/cm^2). Apparently, the variation trend of the d_{33} values was analogous to that of P_r, suggesting that the enhancement of the d_{33} may be attributed to the improvement of the P_r. It has been revealed that the switching current peak is an effective tool to analysis the polarisation switching behaviour, and evaluate the ferroelectric properties of ceramics [38]. *I-E* curves of the Mo/Cr doped ceramics are shown in Fig. 5(e) and apparent ferroelectric switching peaks can be observed. The detailed ferroelectric switching peaks values *vs* doping levels are exhibited in Fig. 5(f). We could clearly see that the switching current peaks become higher with an increase in x content when $x < 0.07$, demonstrating that more domains switch, which may be responsible for the P_r variation.

To further elucidate the mechanism underlying the variation in the piezoelectricity and ferroelectricity, Raman spectra of the Mo/Cr doped ceramics are given in Fig. 6(a). Apparent phonon modes could be observed at 186, 206, 590 and 820 cm^{-1} for all the samples. The mode at 186 cm^{-1} (v_1) could be assigned to the vibration of Ca^{2+} ions in the perovskite-like layers and the mode located at about 206 cm^{-1} (v_2) could be attributed to the interbond angle bending vibration of NbO_6 octahedra. While the modes located at 590 cm^{-1} (v_3) and 820 cm^{-1} (v_4) can be ascribed to the asymmetric and symmetric vibrations of (Nb, Mo/Cr)–O bond, respectively [39–41]. The mode at 590 cm^{-1} exhibited a visible high-frequency shift, as shown in Fig. 6(b). This phenomenon may be due to the smaller ions radii of Mo/Cr compared with that of Nb ion, which results in an increase in the binding strength of the (Nb, Mo/Cr)–O bond [42]. On the other hand, the normalized intensity of 206 cm^{-1} peak in respect of the 186 cm^{-1} peak is shown in Fig. 6(c). From the figure, one could see that the variation of normalized intensity was opposite to that of P_r and d_{33}, demonstrating that the enhancement of piezoelectricity and ferroelectricity may be related to the distortion of NbO_6 octahedron [40].

Thermal stability is an important indicator of the material for high temperature application. To verify the thermal stability of the Mo/Cr co-doped CBN ceramics, the corresponding thermal depoling behaviors of d_{33} were tested, as shown in Fig. 7(a). The ceramics with composition of $x \leq 0.05$ showed no obvious variation in d_{33} values when the depoling temperature was below the T_C, while a slight decrease step could be observed at about 500 °C for the ceramics with $x > 0.05$. When the temperature was raised beyond the T_C, the d_{33} values of all samples dropped very fast and finally became almost zero. It should be noted that d_{33} value of pure CBN did not become zero after annealing at 950 °C, which possessed a T_C value of 943 °C. Similar phenomenon has also been observed in previous reports [43–45]. It might be ascribed to the pinning effect induced by the higher oxygen vacancies concentration, which restricted the reverse switch of the domain during the thermal depoling [31]. Notably, the d_{33} value of the ceramic with $x = 0.07$ remains 14 pC/N after thermal annealing at 850 °C for 2 h, which was 93% of its initial value. Correlation between the d_{33} and d_{33T}/d_{33RT} of previously reported CBN ceramics is shown in Fig. 7(b) [11,13,26,30–35,46]. One can see that the Mo/Cr doped ceramic exhibited a remarkable thermal stability after annealed at 850 °C for 2 h together with a relatively high d_{33} value among the reported CBN-based ceramics. To further determine the thermal behaviors of the samples, the k_p (electromechanical coupling factor) variation *vs* measured temperatures of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics are shown in Fig. 7(c). At room temperature, the k_p value of $x = 0.07$ ceramic (9.4%) was higher than the pristine ceramic (5%), which might be due to the higher d_{33}. Moreover, as the temperature increased, k_p gradually increased. The k_p value of $x = 0.07$ ceramic was 15.2% at 600 °C, still higher than that of pure CBN (7.7%). Good thermal stability along with relatively high k_p in high temperature range confirmed that Mo/Cr doped ceramics might be used in high-temperature range.

The direct current (*dc*) resistivity is a crucial figure of merit for the

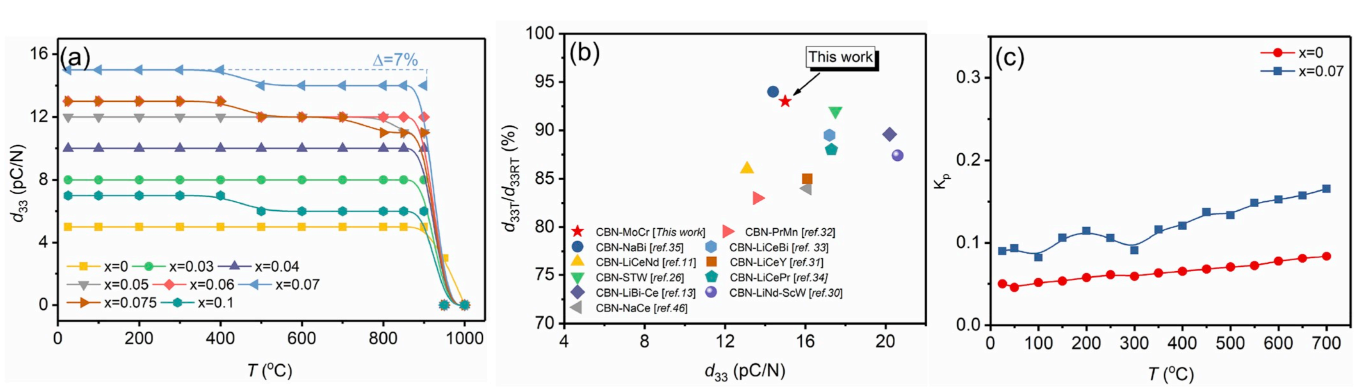

Fig. 7. (a) The effect of thermal depoling on the change of d_{33} after annealing for 2 h. (b) Correlation between the d_{33} and d_{33T}/d_{33RT} of previously reported CBN ceramics. (c) Resonance frequency f_s、f_p and planar electromechanical coupling factor k_p *vs.* Temperature of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics.

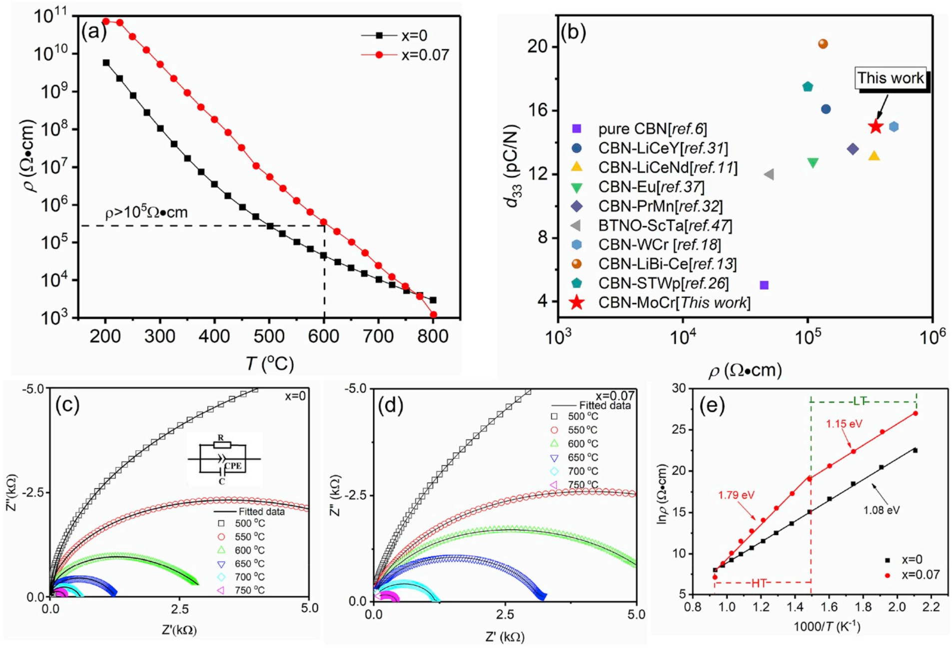

Fig. 8. (a) Variation of the resistivity of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics from 200 °C to 800 °C. (b) Comparison of d_{33} and ρ at 600 °C of CBN-based and BTNO-based ceramics derived from the previously published work and those in this work. (c) and (d) Cole-Cole plots of impedance for $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics measured at various temperatures, respectively. (e) logarithmic resistivities *vs.* temperature of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics.

practical applications of piezoelectric ceramics. As shown in Fig. 8(a), the resistivity of the samples declined as the measured temperature rose, showing a typical thermal activation behavior. Obviously, the resistivity of the ceramic with $x = 0.07$ was higher than the undoped one below 700 °C, indicating that Mo/Cr doping can improve resistivity effectively. When the temperature was beyond 700 °C, the resistivity of the ceramics merged together. The Mo/Cr doped ceramic ($x = 0.07$) with the highest d_{33} value presented a remarkable *dc* resistivity of 3.33×10^5 Ω cm at 600 °C, which was almost 1 order of magnitude larger than that of the pristine CBN (4.48×10^4 Ω cm). In Fig. 8(b), a comparison of d_{33} and ρ at 600 °C of CBN-based and BTNO-based ceramics derived from the previously published work are displayed [11,13,18,26,31,32,37,47]. Obviously, the Mo/Cr doped ceramic exhibited a relatively high d_{33} accompanied by a high resistivity, which can meet the harsh requirements for high temperature piezoelectric applications.

The Cole-Cole plots of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics at varied temperatures are given in Fig. 8(c) and (d) to further study the mechanism for the enhancement of ρ. Both pristine and $x = 0.07$ Mo/Cr doped ceramics exhibited single semicircular plot at different temperature, suggesting the conduction behavior of both ceramics were dominated by grain [48]. A model of a parallel equivalent circuit (R-CPE-C) was established to fit the resistance using Z-view software, as shown in the *inset* of Fig. 8(c). The impedance plots of both the samples fitted well with the experimental data. Logarithmic resistivity *vs* temperature of $x = 0$ and $x = 0.07$ Mo/Cr doped ceramics obtained from the *dc* resistivity is shown in Fig. 8(e). As can be seen, the relationship of lnρ with 1/T can be fitted by one straight line for the undoped ceramic, indicating there may be only one conduction mechanism in the whole tested temperature range. The activation energy (E_a) of ρ is given in Fig. 8(e), which was calculated by the Arrhenius equation:

$$\rho = Aexp(E_a/kT) \quad (2)$$

where E_a is the activation energy of the charge carriers, k is the Boltzmann constant and T is the absolute temperature. The E_a of the pristine CBN was 1.08 eV, indicating that the electrical conduction process was basically dominated by the motion of oxygen vacancies in the whole temperature region [49]. While for the Mo/Cr doped ceramic with $x = 0.07$, the resistivity plot cannot be fitted through a straight line, demonstrating that there should be two different conduction mechanism in the whole temperature range. At low temperature region, the E_a was calculated to be 1.15 eV, which was higher than the pristine one, suggesting that Mo/Cr substitution could reduce the concentration of oxygen vacancies effectively. At high temperature region, the obtained E_a was 1.79 eV, which agreed well with half of the band gap energy E_g of CBN, confirming the intrinsic conduction behaviour in the high-temperature region [11]. In general, the conduction process in the low temperature region (LT) could be attributed to the oxygen vacancy conduction, while intrinsic conduction dominated in the high temperature region (HT) for the Mo/Cr doped ceramics [46,50,51]. All these results demonstrated that Mo/Cr doping could effectively reduce the concentration of oxygen vacancies, which would be beneficial to the enhancement of *dc* resistivity.

The mechanism for the reduction of oxygen vacancies can be analysed by electrochemical means. During the sintering process, the Bi_2O_3 volatilized and oxygen vacancies would be generated to satisfy the charge neutrality [51–53] (Eq. (3)). It was the most important mechanism of oxygen vacancy generation in CBN ceramics, which resulted in the *p*-type conduction behavior of CBN ceramics, expressed by Eq. (4). After Mo/Cr doping, Mo^{6+} substituting for Nb^{5+} generated $Mo_{Nb}^{\bullet}$ and one free electron (Eq. (5)) to comply the principles of charge compensation. The donor dopant of Mo^{6+} could eliminate the oxygen vacancies (Eq. (6)) and neutralize the holes using electrons (Eq. (7), which should be partially responsible for the enhancement of *dc* resistivity. On the other hand, Cr^{3+} ions substituting for Nb^{5+} ions produced two lattice defect ($Cr_{Nb}^{''}$) and two oxygen vacancy (Eq. (8)). Generally, doping of low valence cations would reduce the resistivity. However, recent researches had shown that those oxygen vacancies should be restricted around defect central by forming strong combined defect dipole, which might not participate in the conductive process and could be treated as extrinsic oxygen vacancy ($V_{oe}^{\bullet\bullet}$) [52,54]. However, as the concentration of extrinsic oxygen continued increase, the concentration of intrinsic oxygen vacancies would be decreased to keep the equilibrium concentration of oxygen vacancies [55]. Similar defect dipole ($Cr_{Nb}^{''} - V_{oe}^{\bullet\bullet}$) might also be formed in the Mo/Cr co-doped CBN ceramics, expressed by

Eq. (9). Based on those analysis, the enhancement in resistivity of Mo/Cr doped ceramics was reasonable. In addition, lower oxygen vacancies may also account for the decrease in dielectric loss at high temperatures, as shown in Fig. 4(b).

$$Bi_2O_3 \rightarrow 2\mathrm{Bi(gas)} + \frac{3}{2}O_2(\mathrm{gas})\uparrow + 2V_{Bi}^{'''} + 3V_o^{\bullet\bullet} \tag{3}$$

$$V_o^{\bullet\bullet} + \frac{1}{2}O_2 \rightarrow O_o^x + 2h^{\bullet} \tag{4}$$

$$2MoO_3 \overset{Nb_2O_5}{\rightarrow} 2Mo_{Nb}^{\bullet} + 4O_o^x + O_2\uparrow + 2e^{'} \tag{5}$$

$$2MoO_3 + V_o^{\bullet\bullet} \overset{Nb_2O_5}{\rightarrow} 2Mo_{Nb}^{\bullet} + 6O_o^x \tag{6}$$

$$h^{\bullet} + e^{'} \rightarrow null \tag{7}$$

$$Cr_2O_3 \overset{Nb_2O_5}{\rightarrow} 2Cr_{Nb}^{''} + 2V_{oe}^{\bullet\bullet} + 3O_o^x \tag{8}$$

$$Cr_{Nb}^{''} + V_{oe}^{\bullet\bullet} \rightarrow \left(Cr_{Nb}^{''} - V_{oe}^{\bullet\bullet}\right) \tag{9}$$

4. Conclusions

Mixed valence cations Mo/Cr modified $CaBi_2Nb_2O_9$ ceramics were prepared using the conventional solid-state sintering method. Mo/Cr co-doped in B site reduced oxygen vacancies remarkably and enhanced the tetragonality of lattice structure, thus improved the piezoelectric properties, thermal stability and resistivity of the ceramics. Especially, the ceramic with a composition of $x = 0.07$ exhibited an optimized piezoelectricity with a high d_{33} of ~15 pC/N and an ultra-high T_C of ~939 °C. Moreover, an excellent thermal stability performance that d_{33} remained 93% of its original value even after being annealed at 850 °C for 2 h and an excellent electrical resistivity of 3.33×10^5 Ω•cm at 600 °C were also achieved in this ceramic. This study further proved that B-site mixed valence cations co-doping can be employed to tailor piezoelectric properties of BLSFs, in addition to manipulating the phase boundary.

Conflicts of interest

The authors have no conflicts to declare.

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